

GRAYLING AAF, MI, USA

WMO: 722093

Lat: 44.680N

Lon: 84.729W

Elev: 1149

StdP: 14.10

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.1	-3.7	-17.2	2.2	-10.6	-9.0	3.5	-2.1	23.7	23.4	20.7	24.2	0.5	320	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.4	88.2	69.9	85.0	68.2	81.5	65.9	73.0	81.8	71.3	79.4	69.6	76.7	8.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	117.1	75.6	68.9	110.8	74.6	67.5	105.5	73.1	37.3	81.9	35.8	78.8	34.4	77.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	16.9	15.3	-22.5	92.3	7.5	2.9	-27.9	94.4	-32.3	96.1	-36.5	97.8	-41.9	99.9		
			-22.9	76.2	7.4	2.1	-28.2	77.7	-32.5	78.9	-36.7	80.0	-42.0	81.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	19.5	18.8	28.2	41.0	53.7	62.2	66.7	64.9	58.4	46.4	34.4	24.9
	DBStd	19.50	11.28	11.55	12.51	9.50	9.23	7.17	6.46	6.13	8.00	8.62	9.42	9.66
	HDD50	4313	946	874	683	299	64	3	0	0	14	179	472	779
	HDD65	8182	1411	1294	1141	720	370	138	58	79	229	581	917	1244
	CDD50	1903	0	0	8	29	180	368	519	461	266	67	5	0
	CDD65	293	0	0	0	0	20	53	112	75	30	3	0	0
	CDH74	3673	0	0	10	21	352	669	1357	873	357	34	0	0
	CDH80	981	0	0	1	0	81	173	414	220	88	4	0	0
	WSAvg	6.0	6.4	6.9	6.6	7.5	6.6	5.5	4.8	4.7	4.8	5.8	6.2	6.5
	PrecAvg	33.5	2.1	1.5	1.7	3.0	3.2	4.0	3.4	3.1	3.2	3.3	2.8	2.0
(15) Precipitation	PrecMax	41.4	3.3	2.7	4.3	4.5	5.8	8.7	8.1	5.4	4.9	9.2	5.8	4.4
	PrecMin	27.0	0.7	0.5	0.5	0.9	0.7	1.5	1.4	1.4	0.9	0.8	1.1	0.6
	PrecStd	4.0	0.8	0.6	0.9	1.1	1.3	1.7	1.5	1.1	1.1	1.9	1.4	0.9
	PrecStd	4.0	0.8	0.6	0.9	1.1	1.3	1.7	1.5	1.1	1.1	1.9	1.4	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	47.8	51.7	72.3	77.2	86.9	89.9	92.4	89.9	88.0	79.0	66.5	53.4	(19)
	0.4% MCWB	44.9	46.3	58.7	57.4	67.4	71.8	71.6	71.1	69.0	63.9	55.1	50.7	(20)
	2% DB	42.1	42.7	59.0	71.8	81.8	86.0	88.5	86.1	81.8	73.0	57.3	44.7	(21)
	2% MCWB	39.6	38.6	50.1	54.9	62.9	67.8	70.3	69.2	65.5	61.3	50.1	42.3	(22)
	5% DB	37.1	37.2	52.5	66.1	78.4	81.2	84.6	81.7	78.3	67.9	53.9	40.0	(23)
	5% MCWB	35.4	34.4	44.9	52.0	61.4	65.0	68.9	66.9	64.6	59.8	48.9	38.2	(24)
	10% DB	34.6	34.2	46.2	59.4	73.1	77.7	81.2	79.1	74.0	61.8	48.4	36.5	(25)
	10% MCWB	33.3	32.0	40.1	47.6	58.3	62.8	66.7	65.2	63.3	55.6	44.6	35.1	(26)
	0.4% WB	46.3	46.1	59.7	61.8	70.5	73.2	76.1	74.3	72.9	68.3	57.1	50.9	(27)
	0.4% MCDB	46.8	49.3	71.7	70.4	82.0	85.4	86.2	82.0	81.8	75.8	63.3	52.5	(28)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	39.7	39.2	52.6	57.0	67.4	70.5	73.5	72.2	70.0	64.8	52.6	42.0	(29)
	2% MCDB	41.9	42.8	57.6	67.5	76.7	80.1	82.7	80.1	75.7	69.6	56.7	43.0	(30)
	5% WB	36.0	35.2	45.7	53.3	64.4	68.4	71.7	70.4	68.0	60.4	48.6	38.1	(31)
	5% MCDB	37.2	37.4	50.1	62.9	73.6	76.0	79.4	77.6	73.9	65.0	52.8	39.5	(32)
	10% WB	33.3	32.3	40.3	49.0	61.0	66.8	69.9	68.5	65.8	56.3	45.4	35.3	(33)
	10% MCDB	34.3	33.8	46.2	58.3	69.9	74.1	77.3	75.2	72.6	61.3	48.4	36.4	(34)
(35) Mean Daily Temperature Range	MDBR	15.3	18.8	22.5	23.7	26.4	25.5	25.4	24.8	24.6	19.7	15.5	12.1	(35)
	5% DB	14.1	19.4	28.1	34.6	32.5	29.9	28.6	28.4	29.0	28.9	22.8	15.2	(36)
	5% WB	12.3	15.6	18.7	19.8	16.1	14.7	13.3	14.0	15.3	18.1	16.7	13.1	(37)
	MCDBR	12.9	17.6	24.3	29.7	26.1	22.8	22.3	21.9	21.6	23.1	20.7	14.1	(38)
	MCWBR	11.9	15.0	18.0	18.8	14.6	12.7	12.1	12.1	13.3	16.8	16.5	12.8	(39)
	MCWBR	11.9	15.0	18.0	18.8	14.6	12.7	12.1	12.1	13.3	16.8	16.5	12.8	(39)
(40) Clear-Sky Solar Irradiance	taub	0.277	0.294	0.311	0.342	0.372	0.392	0.390	0.379	0.366	0.330	0.298	0.272	(40)
	taud	2.346	2.285	2.412	2.378	2.355	2.318	2.340	2.363	2.400	2.498	2.522	2.433	(41)
	Ebn at Noon	267	283	294	292	285	278	277	277	271	268	261	260	(42)
	Edh at Noon	24	31	32	36	38	40	38	36	32	25	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg	401	692	1132	1439	1729	1898	1919	1630	1284	767	433	305	(44)
	RadStd	37	77	132	136	158	126	91	97	119	90	61	31	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page